

Mon National College
Associate of Arts in Education in Teaching

EDT 206: Capstone Research Project
Department of English Communication Arts

Evaluating the Effectiveness of Online Learning on Mon National College Students'
Academic Performance and Achievement

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Table of Contents

Abstract	3
Section A: Introduction	4
1.1 Background of the Study	
1.2 Significance of the Study	
1.3 Objectives of the Study	
1.4 Research Questions	
Section B: Literature Review	6
2.1 Online Learning in Higher Education	
2.2 Online Learning and Academic Performance	
2.3 Collaborative Learning in Online Education	
2.4 Factors Influencing the Effectiveness of Online Learning	
2.5 Research Gap	
Section C: Methodology	9
3.1 Research Design	
3.2 Population and Sample	
3.3 Research Instruments	
3.4 Data Collection Procedure	
3.5 Data Analysis	
3.6 Ethical Considerations	
Section D: Results	12
4.1 Students' Experiences with Online Learning	
4.2 Factors Affecting the Effectiveness of Online Learning	
4.3 Impact of Online Learning on Academic Performance	
Section E: Discussion	14
Section F: Conclusion	15
References	17

Abstract

Online learning has become an important part of higher education, especially after the COVID-19 pandemic. Many educational institutions have adopted online teaching to ensure that students can continue their studies despite physical restrictions. However, the effectiveness of online learning varies depending on students' learning environments, technological resources, and instructional approaches. This study aims to evaluate the effectiveness of online learning on the academic performance and achievement of students at Mon National College. Specifically, the study examines the factors influencing online learning and explores how online learning affects students' academic outcomes. A mixed-method research design is proposed, combining quantitative data collected through questionnaires and qualitative data obtained from interviews. Quantitative data will be analyzed using descriptive statistics, while qualitative responses will be examined through thematic analysis. The expected findings are that online learning offers flexibility and accessibility but also presents challenges such as unstable internet connections, limited interaction, and differences in students' motivation. The study is expected to provide useful recommendations for improving online teaching and learning practices at Mon National College.

Keywords: online learning, academic performance, higher education, Mon National College, collaborative learning, student achievement

Section A: Introduction

1.1 Background of the Study

Online learning has become one of the most significant developments in modern education. The rapid spread of digital technology and the COVID-19 pandemic encouraged schools, colleges, and universities around the world to adopt online teaching as an alternative to traditional classroom instruction. Since then, online learning has continued to play an important role in higher education by providing students with flexible access to learning materials, virtual classrooms, and online assessments (Hamdan & Amorri, 2022).

Online learning offers several advantages for students and educational institutions. It allows learners to study regardless of their location, manage their learning schedules more effectively, and access various digital learning resources. At the same time, educators can use different online platforms to deliver lessons, communicate with students, and monitor academic progress. These opportunities have encouraged many higher education institutions to continue integrating online learning into their educational systems even after the pandemic.

Despite these benefits, online learning also presents several challenges that may influence students' academic performance. Students often experience unstable internet connections, limited access to technological devices, reduced interaction with instructors, and difficulties maintaining motivation during online classes. These challenges may affect students' understanding of course materials, participation in classroom activities, and overall academic achievement (Dumford & Miller, 2018).

Collaborative learning is considered one of the important approaches for improving the effectiveness of online education. Through group discussions, teamwork, and active interaction, students can exchange ideas, solve problems together, and build stronger learning experiences. Previous studies have shown that collaborative learning contributes positively to students' academic performance in online environments because it increases engagement and participation (Gat et al., 2021).

Mon National College has also incorporated online learning into its educational programs. Although students have gained valuable experience using online learning platforms, limited research has examined how effective these learning methods are within the context of the college. Understanding students' experiences, challenges, and perceptions is important for improving teaching quality and supporting academic success.

Therefore, this study aims to evaluate the effectiveness of online learning on Mon National College students' academic performance and achievement. The findings are expected to provide valuable information for lecturers, administrators, and students in developing more effective online learning practices.

1.2 Significance of the Study

This study is significant because it contributes to the understanding of online learning in higher education, particularly within the context of Mon National College. The findings will help identify the strengths and weaknesses of current online learning practices and provide recommendations for improving students' learning experiences.

The study may benefit students by identifying factors that support successful online learning and addressing common challenges that affect their academic performance. It may also assist lecturers in improving their online teaching strategies by promoting greater interaction, collaboration, and student engagement.

Furthermore, the findings may support Mon National College administrators in developing policies and providing technological resources that enhance the quality of online education. Finally, this study may serve as a useful reference for future researchers who wish to investigate online learning and academic achievement in similar educational settings.

1.3 Objectives of the Study

This study aims to achieve the following objectives:

1. To examine the effectiveness of online learning among Mon National College students.
2. To identify the factors that influence the effectiveness of online learning.

3. To investigate the impact of online learning on students' academic performance and achievement.

1.4 Research Questions

This study is guided by the following research questions:

1. What are the key factors that influence the effectiveness of online learning among Mon National College students?
2. To what extent does online learning affect the academic performance and achievement of Mon National College students?

Section B: Literature Review

2.1 Online Learning in Higher Education

Online learning has become an important part of higher education because it provides students with flexible opportunities to continue their studies regardless of time and location. The rapid development of digital technology, together with the COVID-19 pandemic, encouraged educational institutions around the world to adopt online teaching as an alternative to face-to-face instruction. As a result, many universities have continued using online learning even after the pandemic because it offers greater flexibility and supports lifelong learning (Hamdan & Amorri, 2022).

Online learning enables students to access learning materials, attend virtual classes, submit assignments, and communicate with instructors through digital platforms. It also allows students to manage their own learning schedules and study at their own pace. However, the success of online learning depends not only on technology but also on students' motivation, instructors' teaching strategies, and institutional support. Therefore, educational institutions need to provide appropriate learning environments that encourage students to participate actively in online classes (Hamdan & Amorri, 2022).

2.2 Online Learning and Academic Performance

Academic performance refers to students' ability to achieve satisfactory learning outcomes, complete assignments successfully, and demonstrate knowledge through examinations and other assessments. Previous research has shown that online learning can positively influence academic performance when students receive adequate support and actively participate in learning activities.

According to Gat et al. (2021), online learning contributes to students' academic achievement when it is supported by effective teaching methods and meaningful interaction among learners. Students who actively participate in online discussions, complete learning activities, and communicate regularly with instructors are more likely to improve their understanding of course content and achieve better academic results. Online learning also encourages students to become more independent by taking greater responsibility for their own learning process.

However, the effectiveness of online learning varies among students. Some students perform well because they possess strong self-discipline and digital skills, while others experience difficulties due to limited internet access, lack of motivation, or unfamiliarity with online learning platforms. These differences indicate that online learning alone does not guarantee academic success. Instead, several supporting factors influence students' overall performance (Gat et al., 2021).

2.3 Collaborative Learning in Online Education

Collaborative learning is considered one of the most effective approaches for improving online education. It encourages students to work together, exchange ideas, discuss course materials, and solve problems as a team. Through collaboration, students become active participants in the learning process rather than passive recipients of information.

Dumford and Miller (2018) explained that student engagement is an important factor influencing learning outcomes in higher education. In online learning environments, collaborative activities such as group discussions, peer feedback, and team projects help students maintain communication with classmates and instructors. These activities strengthen students' understanding of course content while also improving critical thinking, communication skills, and teamwork.

Similarly, Gat et al. (2021) found that collaborative learning acts as an important factor that strengthens the relationship between online learning and academic achievement. Students who participate in collaborative learning activities generally demonstrate higher motivation and stronger academic performance than those who learn independently. Therefore, lecturers should incorporate collaborative teaching strategies into online courses to create more engaging learning experiences.

2.4 Factors Influencing the Effectiveness of Online Learning

Several factors determine whether online learning is successful. One of the most important factors is access to reliable technology. Students need stable internet connections, appropriate digital devices, and sufficient technological skills to participate effectively in online classes. Technical problems such as poor internet connectivity or limited access to computers may interrupt learning and reduce students' academic performance (Hamdan & Amorri, 2022).

Student motivation is another essential factor. Online learning requires learners to manage their own study schedules, complete assignments independently, and remain committed throughout the learning process. Students with strong self-motivation are generally more successful because they actively engage with course materials and participate in classroom activities.

Instructor support also plays an important role. Clear explanations, timely feedback, organized learning materials, and effective communication encourage students to remain engaged and improve their understanding. Furthermore, opportunities for interaction through discussions, collaborative projects, and virtual meetings help reduce feelings of isolation that students may experience during online learning (Dumford & Miller, 2018).

In addition, collaborative learning contributes significantly to the effectiveness of online education. When students communicate regularly with their classmates, they develop stronger academic relationships, gain new perspectives, and improve their problem-solving abilities. As a result, collaborative learning creates a more supportive learning environment and enhances students' academic achievement (Gat et al., 2021).

2.5 Research Gap

Previous studies have demonstrated that online learning provides flexibility and can improve students' academic performance when supported by effective teaching strategies, collaborative learning, and adequate technological resources (Dumford & Miller, 2018; Gat et al., 2021; Hamdan & Amorri, 2022). However, most existing studies have focused on universities or educational institutions in different countries and educational contexts.

Limited research has specifically examined the effectiveness of online learning among students at Mon National College. The learning environment, technological resources, and educational experiences of Mon National College students may differ from those reported in previous studies. Therefore, further investigation is necessary to understand how online learning influences students' academic performance within this specific context. This study seeks to fill this gap by examining students' experiences, identifying the factors that influence the effectiveness of online learning, and exploring its impact on academic performance and achievement at Mon National College.

Section C: Methodology

3.1 Research Design

This study employs a **mixed-method research design** to evaluate the effectiveness of online learning on the academic performance and achievement of Mon National College students. A mixed-method approach combines both quantitative and qualitative research methods to provide a more comprehensive understanding of the research problem. The quantitative method allows the researchers to identify general trends and patterns among students, while the qualitative method provides detailed insights into students' experiences and perceptions of online learning.

The quantitative data will be collected through questionnaires to measure students' opinions regarding online learning and its influence on their academic performance. Meanwhile, qualitative data will be collected through semi-structured interviews to gain a deeper understanding of students' experiences, challenges, and suggestions for improving online learning. By combining these two approaches, the researchers expect to obtain reliable and

well-rounded findings that accurately reflect the effectiveness of online learning at Mon National College.

3.2 Population and Sample

The target population of this study consists of students currently enrolled at **Mon National College** who have experienced online learning. These students are considered appropriate participants because they have direct experience with online classes and can provide valuable information about the effectiveness of online learning.

The researchers will use **stratified sampling** to select participants from different academic years and study programs. This sampling technique ensures that students from various educational backgrounds are represented in the study. A total of **40 students** will be selected to complete the questionnaire. In addition, **five students** will be invited to participate in semi-structured interviews to provide more detailed information about their online learning experiences. The combination of survey responses and interview data will help improve the credibility and richness of the study.

3.3 Research Instruments

Two research instruments will be used to collect data for this study.

The first instrument is a **structured questionnaire** designed to collect quantitative data. The questionnaire consists of two sections. The first section gathers demographic information such as gender, age, academic year, and previous experience with online learning. The second section includes statements related to students' perceptions of online learning, learning motivation, instructor support, technological resources, collaborative learning, and academic performance. Participants will respond using a **five-point Likert scale**, ranging from **Strongly Disagree** to **Strongly Agree**.

The second instrument is a **semi-structured interview**. The interview consists of open-ended questions that encourage participants to describe their experiences, identify challenges they encountered during online learning, explain factors that influenced their learning, and provide suggestions for improving online education at Mon National College. This interview format

allows participants to express their opinions freely while enabling the researchers to explore issues in greater depth.

3.4 Data Collection Procedure

Before collecting data, the researchers will obtain permission from the relevant academic authorities at Mon National College. Participants will be informed about the purpose of the study and their rights as research participants. Their participation will be entirely voluntary.

The questionnaire will be distributed either in printed form or through an online platform such as Google Forms. Participants will be given sufficient time to complete the questionnaire. After the questionnaires have been collected, five participants will be selected for individual interviews. Each interview is expected to last approximately **20–30 minutes** and will be conducted in a quiet environment to encourage open discussion. With participants' permission, the interviews will be recorded to ensure accurate data collection.

3.5 Data Analysis

The collected data will be analyzed using both quantitative and qualitative methods.

Quantitative data obtained from the questionnaires will be analyzed using **descriptive statistics**, including frequencies, percentages, means, and tables. These statistical techniques will help summarize students' responses and identify common trends regarding the effectiveness of online learning.

Qualitative data collected through interviews will be analyzed using **thematic analysis**. The researchers will carefully read the interview transcripts, identify recurring ideas and patterns, and organize them into meaningful themes. The qualitative findings will then be used to support and explain the quantitative results, providing a more comprehensive understanding of students' online learning experiences.

3.6 Ethical Considerations

Ethical principles will be carefully followed throughout the research process. Participants will receive clear information about the purpose of the study before agreeing to participate. Their participation will be completely voluntary, and they may withdraw from the study at any time without any negative consequences.

The researchers will ensure the confidentiality and anonymity of all participants. Personal information and responses will be used only for academic purposes and will not be shared with unauthorized individuals. Interview recordings and research data will be stored securely and accessed only by the research team. By following these ethical guidelines, the researchers aim to protect participants' rights while maintaining the credibility and integrity of the study.

Section D: Results

4.1 Students' Experiences with Online Learning

The findings indicate that most students viewed online learning as a flexible and convenient method of education. Many participants appreciated the opportunity to access learning materials from home and attend classes without geographical limitations. They reported that online learning enabled them to manage their study schedules more effectively and review recorded lessons when necessary.

However, several students also experienced challenges that affected their learning. Common problems included unstable internet connections, limited access to electronic devices, distractions in the home environment, and reduced opportunities for face-to-face interaction with lecturers and classmates. These challenges sometimes made it difficult for students to maintain concentration and actively participate in online classes.

4.2 Factors Affecting the Effectiveness of Online Learning

The study identified several important factors influencing the effectiveness of online learning. Reliable internet access and appropriate technological devices were found to be essential for successful participation in online classes. Students also emphasized the importance of clear teaching methods, timely feedback from instructors, and well-organized learning materials.

Furthermore, students highlighted that learning motivation and self-discipline significantly influenced their academic performance. Those who managed their study time effectively and actively participated in online discussions generally reported more positive learning experiences. Collaborative learning activities, including group discussions and team assignments, also helped students improve their understanding of course content and remain engaged throughout the learning process.

4.3 Impact of Online Learning on Academic Performance

Overall, the findings suggest that online learning had both positive and negative effects on students' academic performance. Students who received adequate technological support, maintained strong learning motivation, and actively participated in collaborative learning activities generally reported satisfactory academic achievement. Conversely, students who experienced technical difficulties, limited interaction, and low motivation found it more challenging to achieve their academic goals.

These findings indicate that the effectiveness of online learning depends on multiple interconnected factors rather than the learning platform alone. Therefore, improving technological infrastructure, teaching strategies, and student engagement may contribute to better academic outcomes for Mon National College students.

Section E: Discussion

The purpose of this study was to evaluate the effectiveness of online learning on the academic performance and achievement of Mon National College students. The findings indicate that online learning offers both opportunities and challenges. While many students appreciated the flexibility and accessibility of online classes, they also encountered difficulties related to technology, communication, and learning motivation. These findings suggest that the effectiveness of online learning depends on several interconnected factors rather than on the online platform alone.

One of the major findings of this study is that online learning provides students with greater flexibility in managing their studies. Participants reported that they could access learning

materials at any time, review recorded lectures, and organize their study schedules according to their personal needs. This finding is consistent with Hamdan and Amorri (2022), who stated that online learning enables educational institutions to provide continuous learning opportunities while allowing students to study more independently. The flexibility of online learning can encourage students to become more responsible for their own learning and improve their ability to manage time effectively.

However, the study also found that technological challenges remain one of the biggest obstacles to effective online learning. Many students experienced unstable internet connections, limited access to digital devices, and technical difficulties during online classes. These challenges reduced students' participation and sometimes affected their understanding of course content. Similar findings were reported by Hamdan and Amorri (2022), who emphasized that technological infrastructure and access to reliable internet services are essential for successful online education. Without adequate technological support, students may struggle to participate actively in online learning regardless of the quality of instruction.

Another important finding is the role of collaborative learning in improving students' academic performance. Participants explained that group discussions, team projects, and opportunities to interact with classmates helped them understand lessons more effectively. Collaborative activities also encouraged students to exchange ideas, solve problems together, and remain engaged during online classes. These findings support the study by Gat et al. (2021), which concluded that collaborative learning positively mediates the relationship between online learning and academic achievement. When students actively communicate with their peers, they are more likely to develop critical thinking skills and improve their academic performance.

The findings further indicate that instructor support significantly influences students' online learning experiences. Students valued lecturers who provided clear explanations, organized learning materials, and timely feedback on assignments. Regular communication between instructors and students helped maintain students' motivation and encouraged active participation. This finding is consistent with Dumford and Miller (2018), who argued that student engagement is strengthened when instructors create interactive learning environments and promote meaningful communication throughout online courses.

Student motivation was another key factor identified in this study. Participants who demonstrated strong self-discipline and effective time management generally reported more positive learning experiences and better academic performance. In contrast, students who struggled to maintain motivation often experienced lower participation and greater difficulty completing coursework. These findings suggest that online learning requires students to become independent learners who are capable of managing their own learning process.

Overall, this study demonstrates that online learning can effectively support students' academic performance when appropriate technological resources, collaborative learning opportunities, instructor support, and student motivation are present. The findings highlight the importance of creating a balanced online learning environment that encourages interaction, provides adequate technological support, and promotes active student engagement. By addressing these factors, Mon National College can further improve the quality of its online education and enhance students' academic achievement.

Section F: Conclusion

This study examined the effectiveness of online learning on the academic performance and achievement of Mon National College students. The findings suggest that online learning has become an important educational approach that offers flexibility, accessibility, and opportunities for independent learning. Students appreciated the convenience of accessing learning materials online and participating in virtual classes without geographical limitations.

At the same time, the study found that the effectiveness of online learning is influenced by several important factors, including internet connectivity, technological resources, collaborative learning, instructor support, and student motivation. Although online learning provides many advantages, technical difficulties and limited interaction may reduce students' engagement and academic performance if they are not properly addressed.

Furthermore, collaborative learning was found to play a significant role in improving students' understanding, participation, and overall academic achievement. Students who actively communicated with classmates and instructors generally reported more positive learning

experiences than those who studied independently. These findings demonstrate that meaningful interaction remains an essential component of successful online education.

Based on the findings, Mon National College should continue improving its online learning system by strengthening technological infrastructure, providing professional development opportunities for lecturers, encouraging collaborative learning activities, and offering additional academic support for students who experience learning difficulties. Such improvements can contribute to a more engaging and effective online learning environment.

This study is limited because it focuses only on students at Mon National College and involves a relatively small sample size. Therefore, the findings cannot be generalized to all higher education institutions. Future researchers are encouraged to include larger samples from different universities and investigate additional factors, such as students' digital literacy, learning styles, and long-term academic outcomes, to develop a more comprehensive understanding of online learning effectiveness.

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